

Name: V. Pavithra
Reg no :192524274

course code: CSA1007
slot: D

CO 1 ASSESSMENT TOOL 2
Agile Sprint Planning & User Story Workshop

**AUTONOMOUS ELECTRIC VEHICLE CHARGING NETWORK
MANAGEMENT SYSTEM**

1. Project Vision and Stakeholders

Vision: Build an intelligent, reliable, and scalable EV charging network platform that lets drivers find, reserve, and pay for charging effortlessly, while giving operators real-time visibility, automated fault management, and dynamic load balancing across the network.

Objectives:

- Provide a seamless end-to-end experience for locating, reserving, and using charging stations.
- Deliver real-time monitoring and automated fault detection to minimize downtime.
- Implement dynamic load balancing to protect the grid and reduce peak energy costs.
- Automate billing and provide operators with actionable business reports.
- Ensure scalability to support thousands of stations and concurrent sessions.

Key Stakeholders:

EV drivers locate, reserve, and pay for charging through the mobile app; network operators own and manage station assets and track uptime, revenue, and utilization; maintenance technicians respond to fault alerts and perform repairs; fleet customers schedule bulk or recurring charging for vehicle fleets; utility and grid providers supply power and set capacity and peak-demand constraints; the payment gateway provider processes automated billing; the product owner defines and prioritizes the backlog; the Scrum Master facilitates Agile ceremonies and removes impediments; and the development team designs, builds, and tests each sprint increment.

2. Epics and User Stories

Requirements are grouped into four epics, each expressed as user stories in the format “As a <user>, I want <feature>, so that <benefit>.”

- **EPIC-1: User Account & Charging Session Management** — locate stations, reserve slots, and start/stop sessions from the app.
- **EPIC-2: Real-Time Charger & Network Monitoring** — give operators live charger health/status and automated fault alerts.
- **EPIC-3: Dynamic Load Balancing & Grid Management** — optimize power distribution to avoid grid overload and prioritize fleet off-peak charging.
- **EPIC-4: Billing, Payments & Reporting** — automate payment on session completion and generate monthly utilization/revenue reports.

3. Acceptance Criteria (Given–When–Then)

Each user story's acceptance criteria are written in Given–When–Then form. Key examples:

- ❖ US-1.1 (Locate stations): Given location services are on, when the driver opens the finder, then stations within 10 km appear with live availability.
- ❖ US-1.3 (Start/stop session): Given the driver is authenticated at a charger, when they tap Start, then charging begins within 5 seconds; stopping generates an invoice within 30 seconds.
- ❖ US-2.1 (Charger dashboard): Given a charger misses two heartbeats, when this occurs, then it is flagged Offline within 60 seconds; overheating faults auto-remove the charger from service.
- ❖ US-3.1 (Load balancing): Given aggregate demand nears the site's grid limit, when this happens, then lower-priority sessions are throttled in real time and rebalanced within 2 minutes of any limit change.
- ❖ US-4.1 (Auto payment): Given a session ends, when energy usage is calculated, then the saved payment method is charged within 1 minute and a receipt emailed; exhausted retries pause charging privileges.

4. Prioritized Product Backlog

Prioritized using MoSCoW. Must-Have (high business value, no dependency issues): locate nearby stations (5 points), start/stop session from app (8 points, depends on station locator), real-time charger status dashboard (8 points), and automatic session payment (5 points, depends on session control). Should-Have (medium value): advance slot reservation (8 points), automated fault alerts (5 points), and dynamic load balancing (13 points), each building on the Must-Have foundation. Could-Have (lower, optimization-focused): fleet off-peak prioritization (8 points) and monthly utilization/revenue reports (5 points).

Rationale: the Must-Have items form the minimum viable product needed for a driver to complete a charging transaction end-to-end. Should-Have items improve reliability and efficiency, and Could-Have items add business optimization once the core platform is stable.

5. Sprint Backlog (Sprint 1)

The four Must-Have stories were decomposed into tasks across Frontend, Backend, and QA, totalling roughly 80 hours for a 5-person team over a two-week sprint:

- US-1.1 – Station locator: design the map UI with markers/availability legend (Frontend, 8h), build a radius-filtered search API (Backend, 6h), and integrate the real-time availability feed (Frontend, 5h).
- US-1.3 – Session control: implement BLE/QR charger authentication (Backend, 8h), build the start/stop session API and state machine (Backend, 8h), and build the in-app session control screen (Frontend, 5h).
- US-2.1 – Operator dashboard: design monitoring wireframes (UX/Frontend, 5h), implement the charger heartbeat listener (Backend, 8h), and build offline/fault alerting logic (Backend, 6h).
- US-4.1 – Payments: integrate the payment gateway for saved-card billing (Backend, 8h) and build automated invoice/receipt generation (Backend, 5h).
- QA: write and execute test cases covering all Sprint 1 stories (8h).

6. Story Point Estimation

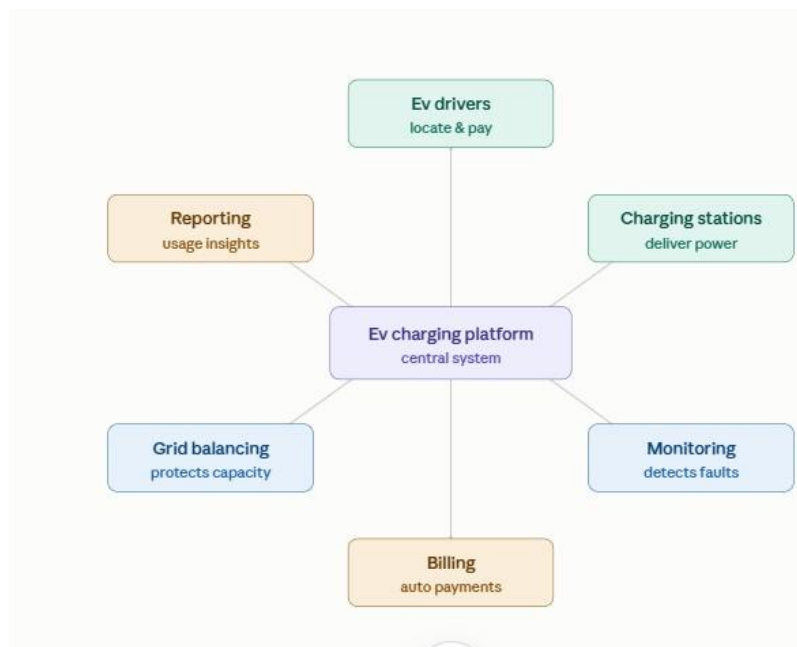
Points were assigned via Planning Poker (Fibonacci scale), reflecting complexity and risk rather than time. Locating stations is rated 5 points as a moderate, low-risk UI and geo-query task.

Advance reservation and start/stop session are each rated 8 points due to reservation-hold timers and hardware-handshake/state-machine complexity respectively. The charger status dashboard is rated 8 points for concurrency and alerting complexity across many devices, while automated fault alerts add only 5 points since they build on that same infrastructure. Dynamic load balancing is the largest item at 13 points, being a safety-critical, hardware-in-loop optimization problem. Fleet off-peak prioritization is rated 8 points for its scheduling and contract-rule logic, and automatic payment and monthly reporting are each rated 5 points as comparatively standard integration and reporting work.

Total backlog size is 65 story points. With a historical team velocity of about 26 points per sprint, the 26 points of Must-Have stories selected for Sprint 1 represent a full, achievable sprint load.

7. Sprint Goal and Expected Deliverables

Sprint Goal: “Enable an EV driver to find a nearby charging station, start and stop a charging session from the mobile app, and be billed automatically while giving network operators real-time visibility into charger status.”



Expected Deliverables:

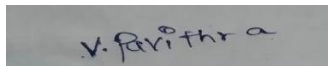
- A functional station-locator map with real-time availability within a chosen radius.

- An end-to-end session flow: authenticate → start → monitor → stop.
- A live operator dashboard showing online/offline/fault status for all chargers.
- Automated payment processing and receipt generation on session completion.
- An automated test suite covering the four Must-Have stories with QA sign-off.
- A working increment deployed to staging, demonstrable at the Sprint Review.

Definition of Done:

- Code peer-reviewed, merged, and passing CI build.
- Unit and integration tests passing, with $\geq 80\%$ coverage on new code.
- Acceptance criteria for each story verified against Given-When-Then conditions.
- Feature deployed and verified in staging, and accepted by the Product Owner.

I V. PAVITHRA (19242524274) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

A rectangular box containing a handwritten signature in dark ink, which appears to read "V. Pavithra".